Project Review-1 Report

Defect Detection & Analytics System for Pharmaceutical Capsules

1. Problem Statement and Objectives

Problem Statement:

Manual inspection of pharmaceutical capsules is a critical quality control process but suffers from significant limitations. Human inspectors are prone to fatigue, leading to inconsistent defect detection rates. Furthermore, high-speed production lines exceed human visual processing capabilities, resulting in potential release of defective batches or high scrap rates due to false rejections.

Motivation:

Ensuring 100% quality in pharmaceutical products is mandatory for patient safety. Automated Visual Inspection (AVI) systems offer a reliable, non-contact, and high-speed alternative. However, traditional AVI systems struggle with subtle defects and require extensive feature engineering.

Objectives:

1. Develop an Automated Visual Inspection system for pharmaceutical capsules using Deep Learning.
2. Implement an Unsupervised Anomaly Detection model (PatchCore) to detect unknown defects with high accuracy.
3. Integrate a Supervised Classifier (ResNet18) to categorize specific defect types (e.g., Crack, Poke, Squeeze).
4. Create a comprehensive Analytics Dashboard for real-time monitoring and reporting.

2. Detailed Literature Review

The field of anomaly detection in manufacturing has evolved significantly. Early approaches relied on traditional Computer Vision techniques such as edge detection and morphological operations (image differencing). These methods are computationally efficient but lack robustness against lighting variations and complex textures.

Supervised Deep Learning (e.g., CNNs like YOLO, ResNet) improved performance but requires large, fully labeled datasets of defined defects. In pharmaceutical manufacturing, defective samples are rare, making supervised training difficult due to class imbalance.

Current State-of-the-Art methodologies focus on Unsupervised Learning, training only on 'Good' samples. Autoencoders and GANs reconstruct images and detect anomalies based on reconstruction error, but often produce blurry outputs. Recent advances like SPADE and PatchCore (Roth et al., 2022) utilize pre-trained feature extractors (e.g., WideResNet) and memory banks of nominal features. PatchCore achieves SOTA performance on the MVTec AD dataset by using coreset sampling to reduce memory footprint while maintaining high detection accuracy. This project adopts PatchCore for its superior performance and low inference latency.

3. Proposed Methodology / System Design

Our approach integrates Unsupervised Anomaly Detection with Supervised Classification to provide a robust inspection solution.

System Architecture:

1. Image Acquisition: High-resolution images of capsules are captured (simulated via upload).
2. Preprocessing: Images are resized to 256x256 and normalized.
3. Anomaly Detection (PatchCore):
   - Features are extracted using a pre-trained Wide ResNet-50 backbone.
   - Local features are compared against a memory bank of 'Good' features using Nearest Neighbor Search.
   - An Anomaly Score is generated. If Score > Threshold, it is flagged as specific Defect.
4. Defect Classification (ResNet-18):
   - Defective images are passed to a secondary classifier trained on specific defect types (Crack, Poke, etc.) to identify the root cause.
5. Analytics & Visualization:
   - Results are logged to an SQLite database.
   - A Streamlit Dashboard visualizes trends, defect rates, and generates PDF reports.

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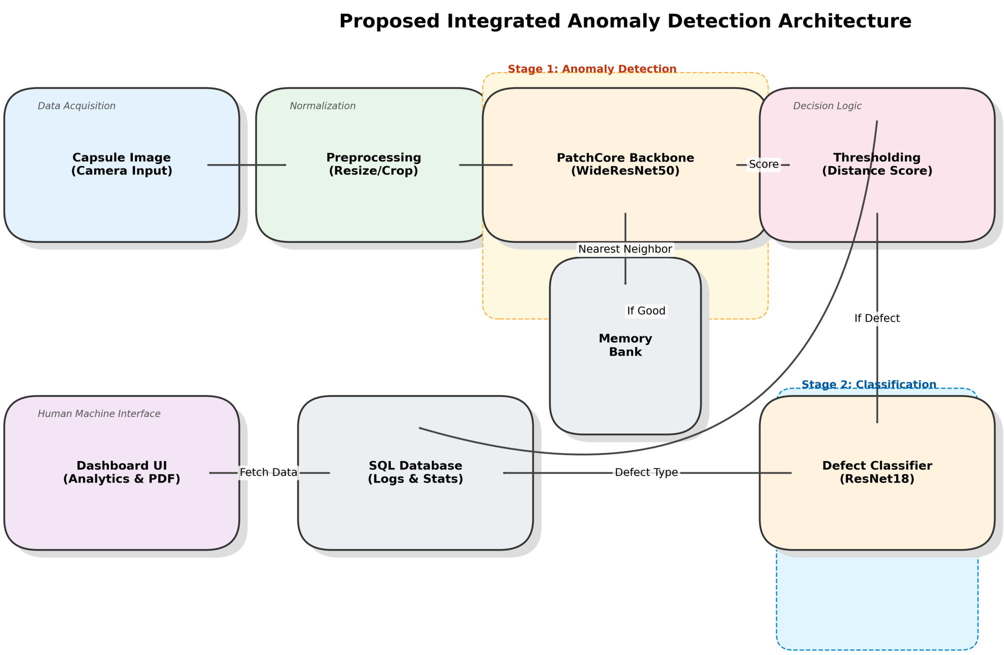

4. Work Plan / Timeline

The project is structured into 5 phases over a 10-week period. We have currently completed Phase 3 (Analytics Module) and are in the Integration & Testing phase.